Elektronik I 1TE624 HT2008

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Kurslitteratur
Electronics: A Systems Approach

Electronics:
A Systems Approach

3rd Edition

by Neil Storey
published by
Prentice Hall

Kursplan

- **Förkunskaper**: Elektromagnetism I. Mätteknik. Transformmetoder förutsätts läsas samtidigt.
  Utbildningsnivå: Grundnivå
  Betygsskala: För denna kurs ges betygen U underkänd, 3 godkänd, 4 icke utan beröm godkänd och 5 med beröm godkänd
- **Huvudområde**
- Teknik
  Mål
- Efter fullgjord kurs skall studenten kunna:
- analysera frekvensgång och fasgång för passiva och aktiva filter och använda bodediagram
- konstruera förstärkare med hjälp av OP-förstärkare
- använda dioder och transistorer i enkla kopplingar
- konstruera kombinatoriska nät och sekvensnät med D-vippor
- använda ett simuleringsprogram för analys och konstruktion av elektroniska kretsar

- **Innehåll**
- Tvåpoler och tvåpolssatsen. Överföringsfunktioner, gränsfrekvens, bodediagram. Operationsförstärkaren, dioden. Fälteffekt- och bipolära transistorn. Digitala grundkretsar, Boolsk algebra, Karnaughdiagram. Analys och syntes av kombinatoriska kretsar och synkrona sekvenskretsar.
- **Undervisning**
- Föreläsningar, lektioner och laborationer
- **Examination**
- Skriftlig tentamen vid kursens slut (4 hp). För godkännande fordras även godkända laborationer (1 hp).

- **<u>Kursen går på halvfart!</u>**

Två områden: Analog och Digital Elektronik

- Kursen innehåller både analog och digital elektronik.
- Grundläggande koncept för respektive område gås igenom med stöd av labbar och räkneövningar
- Period I: Analog elektronik
- Period II: Digital elektronik
- Tenta vid slutet av period II

Planering period I

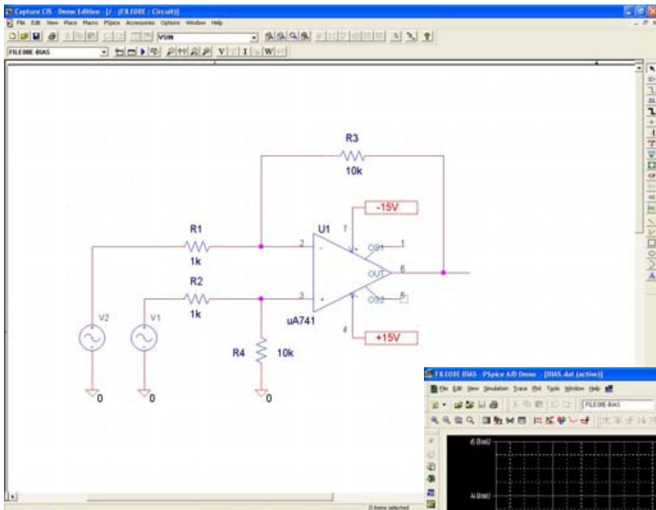
- **Föreläsningar**
- Introduktion, repetition — kap. 1, 3.2-3.4
- Tvåpol, förstärkning, bode — kap. 3.5-3.9
- OP — kap. 5.1-5.7, 4.4 inkluderas i 5.3
- -- “ -- — -- “ --
- Filter, oscillatorer — kap. 10.-10.2, 11.1-11.2.2
- Dioden — kap. 6.4-6.6, (6.7.1-6.7.2 översiktligt), 6.8.1-6.8.2
- fälteffekttransistorn — kap. 7.1-7.4 översiktligt, 7.5-7.6 (valda delar)
- bipolärtransistorn — kap. 8.1-8.4 översiktligt, 8.5-8.6 (valda delar)

- **Labbar**
- R, C, OP, mätningar och simuleringar
- Diod och transistor (+R,C,OP) mätningar och simuleringar

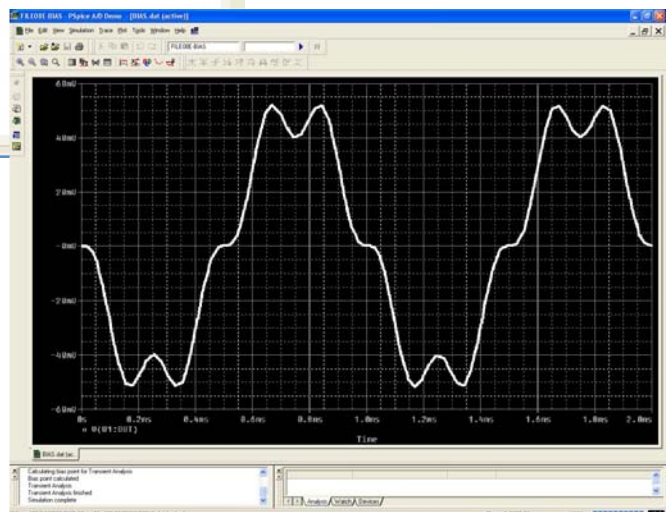
- **Övningar**
- tvåpol, fyrpol, bode, etc
- OP-kretsar
- filter, osc, dioder
- transistorkopplingar

- **Tidsschema**
- F1, F2, F3, Ö1, F4, F5, Ö2, F6, Ö3, Lab1, F7, F8, Lab2, Ö4

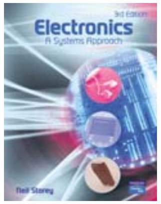
Kretssimulering



OrCAD
PSpice

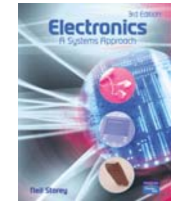


- Simulering ingår i labbar
- Även övnings-simuleringar finns kopplade till texten i boken
- Både simulatorer och övningsfiler finns att komma åt i från hemsidan



Electronic systems

- Introduction
- Electronic systems
- Distortion and noise
- System design.



Introduction

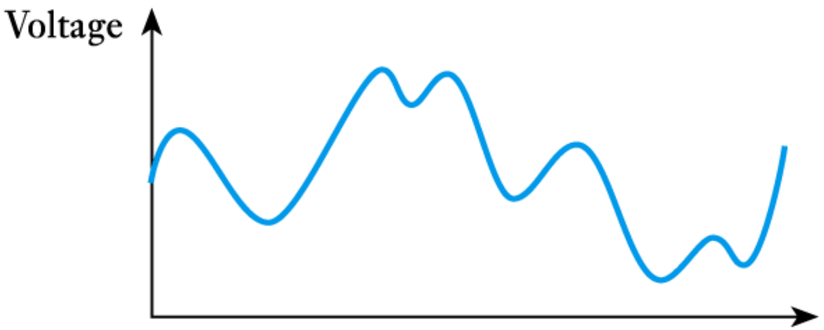
- The world in which we live is constantly changing.
- To survive, we need to respond to changes in our environment.
  - To respond we sense a changing quantity (the **input**).
  - And modify some other quantity (the **output**).
- We often use machines to respond on our behalf.
  - The nature of these machines is that they **sense** some input quantity, **process** the information, and then **control** some output quantity.

Introduction (contd.)

- The world about us is characterised by a number of **physical properties** or **quantities.**
  - e.g. temperature, pressure, humidity, etc.
- Physical quantities may be *continuous* or *discrete*.
- **Continuous quantities** change smoothly and can take an infinite number of values.
- **Discrete quantities** change abruptly from one value to another.
  - Most real-world quantities are continuous.
  - Many man-made quantities are discrete.

Introduction (contd.)

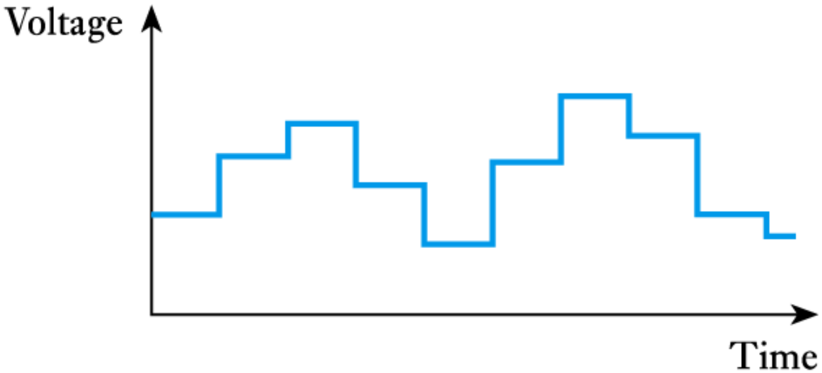
- It is often convenient to represent physical quantities by electrical signals. These can also be continuous or discrete.
- Continuous signals are often described as **analogue.**



(a) An analogue signal

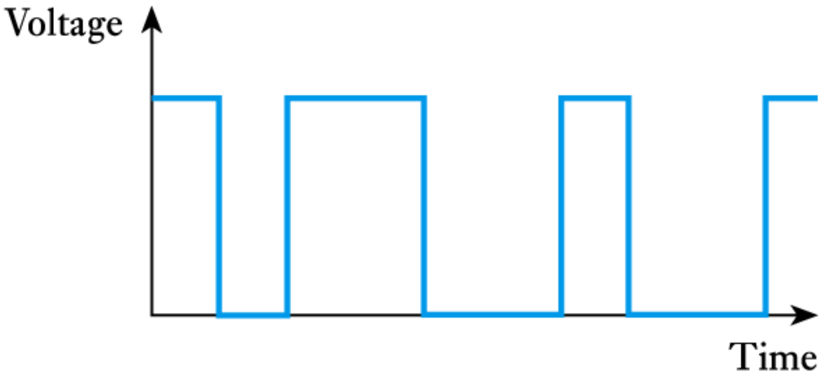
Introduction (contd.)

- Discrete signals are often described as **digital signals.**



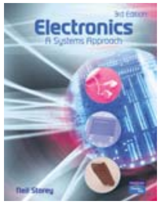
(b) A multi-valued digital signal

- Many digital signals take only two values and are referred to as **binary signals.**



(c) A binary signal

Electronic systems



1.2

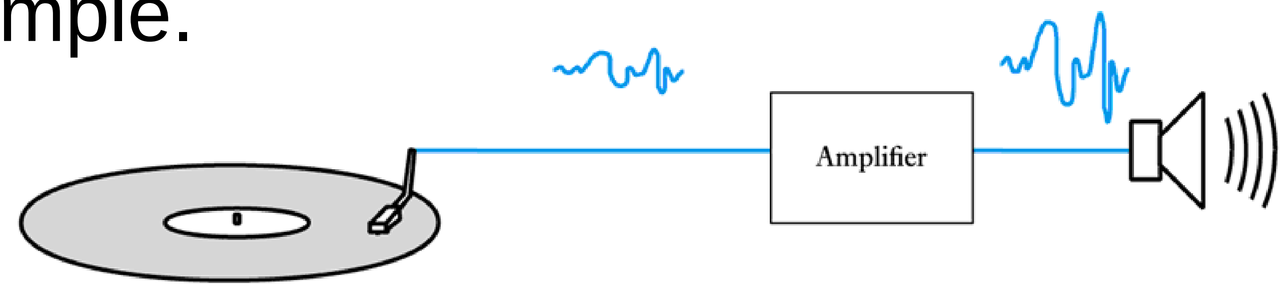
- A system can be defined as

  Any closed volume for which all the inputs and output are known.

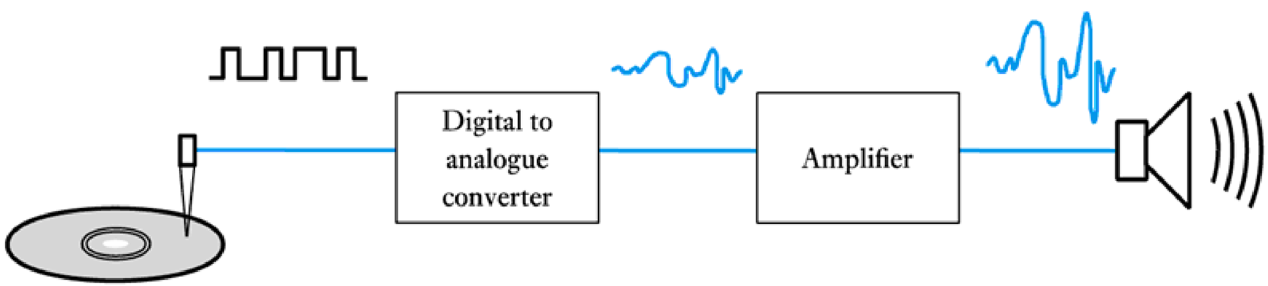
- Examples include:
  - – an engine management system
  - – an automotive system
  - – a transportation system
  - – an ecosystem.
- Inputs and outputs will reflect the nature of the system.

Electronic systems (contd.)

- Electronic systems can take many forms, for example.



(a) A record player

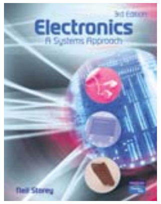


(b) A compact disc player

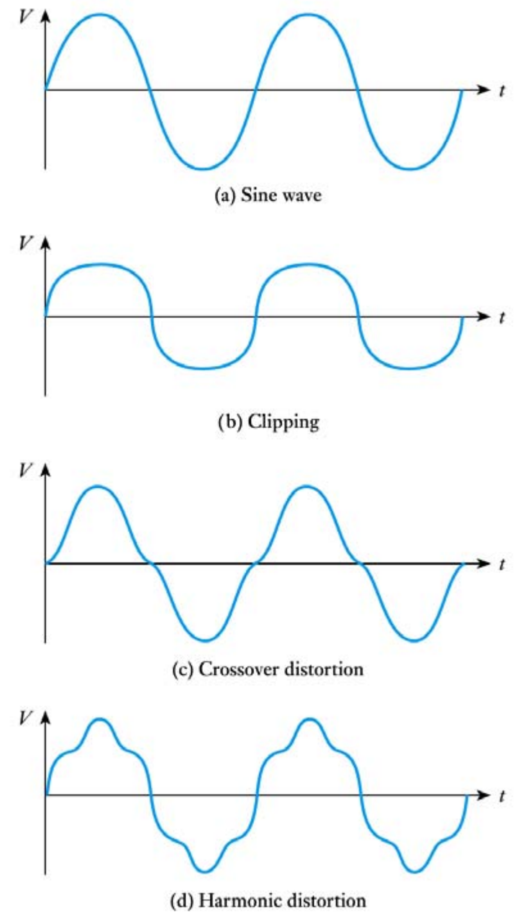
Electronic systems (contd.)

- Components that interact with the outside world are termed **sensors** and **actuators.**
  - In the previous examples the pickup or laser scanner represents a sensor.
  - In the previous examples the loudspeaker represents an actuator.
- We will look at sensors and actuators in more detail in later lectures.

Distortion and noise

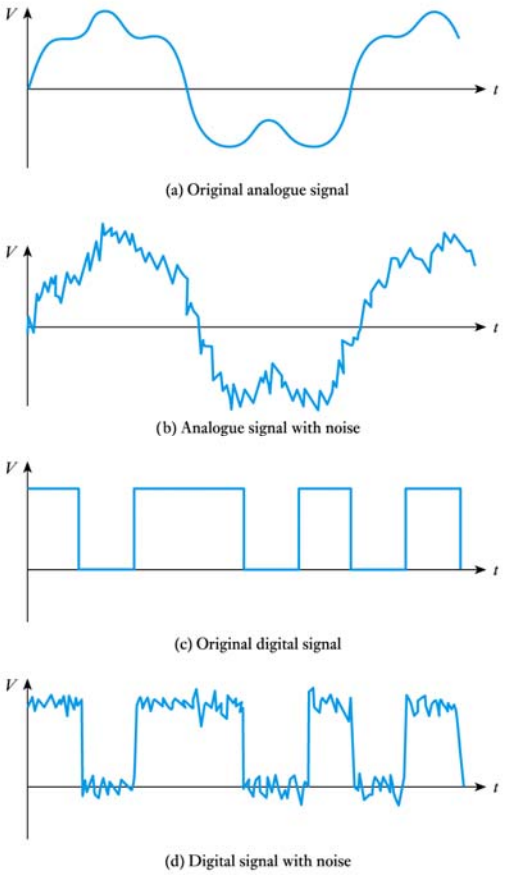


- All systems distort electrical signal to some extent
  - – examples include clipping, crossover distortion and harmonic distortion.
- Distortion is **systematic** and is **repeatable.**

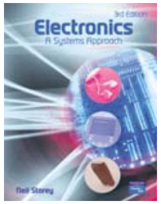


Distortion and noise (contd.)

- All systems also add noise to the signals that pass through them.
- Unlike distortion, noise is **random** and not repeatable.
- Noise can often be removed from digital signals but this is often impossible with analogue signals.



System design



1.4

- The task of designing an electronic system can be simplified by adopting a methodical approach.
- Generally this involves a **top-down approach.**
  - Customer requirements
  - Top-level specification
  - Choice of technology
  - Top-level design
  - Detailed design
  - Module construction and testing
  - System testing.

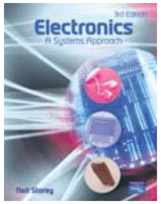
System design (contd.)

- **Electronic design aids**
  - schematic capture
  - circuit simulation
  - PCB or VLSI layout packages.
- Circuit simulation greatly assists our understanding of the operation of a circuit.
  - Common examples include PSpice and Multisim.
  - Simulation **demonstration files** are available to support the material in the course text.

Key points

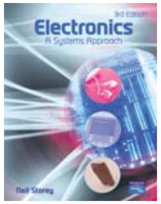
- Systems interact with the world using sensors and actuators.
- Physical quantities can be either continuous or discrete.
- Physical quantities are often represented by signals.
- Useful electronic systems take input signals, process this information and produce appropriate outputs.
- Distortion and noise are always present.
- System design normally follows a top-down approach.
- Electronic design tools, such as simulators, are invaluable.

Laboratory measuring instruments



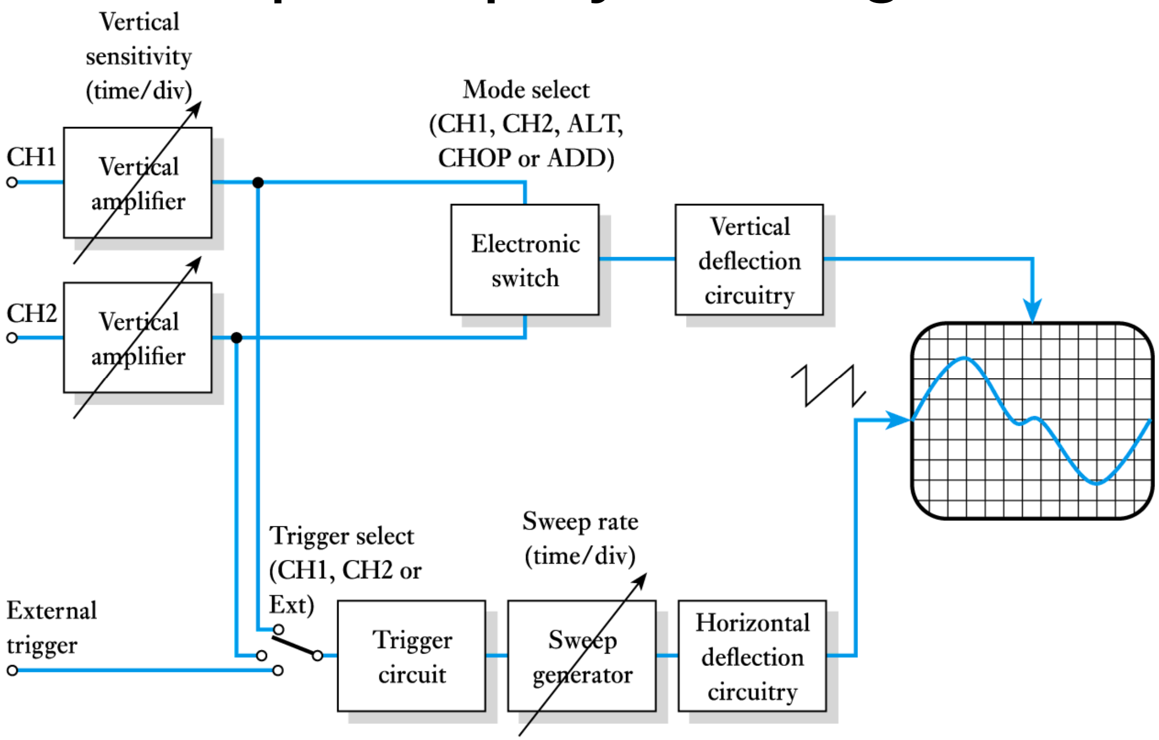
- Often the object of sensing a physical quantity is to **measure** it.
- Here we will look at three forms of measuring instrument:
  - analogue oscilloscope
  - digital oscilloscope
  - digital multimeter.

Analogue oscilloscope



2.5.1

- An oscilloscope displays voltage waveforms.



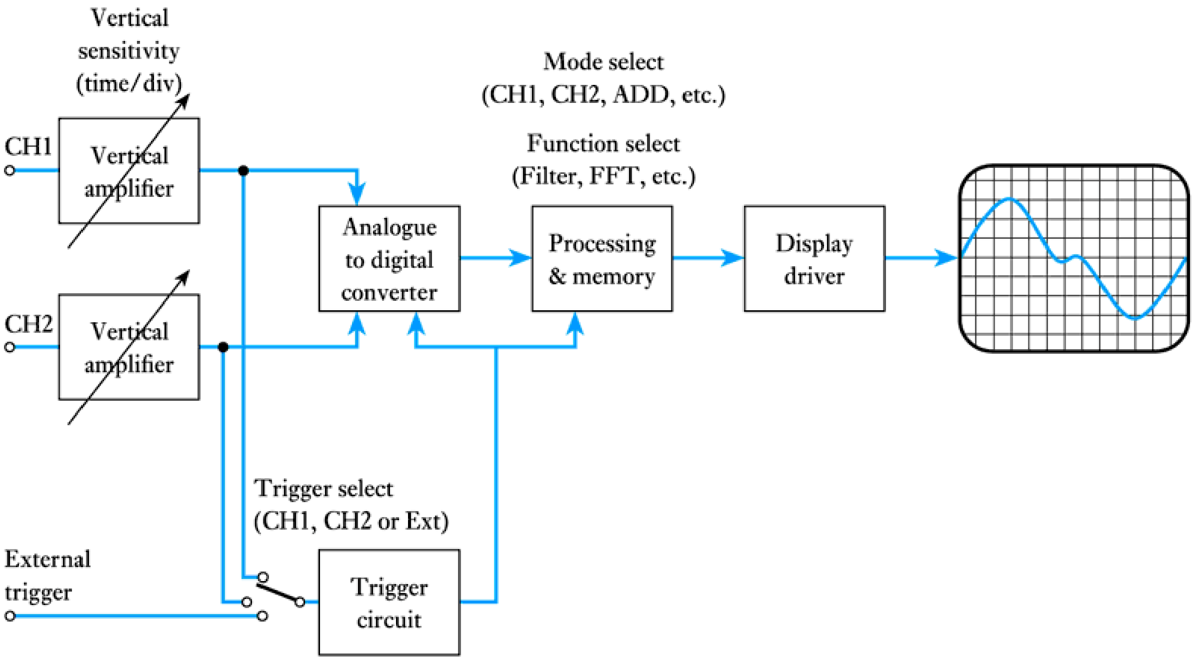
A simplified block diagram

Analogue oscilloscope (contd.)

- A typical analogue oscilloscope

Digital oscilloscope

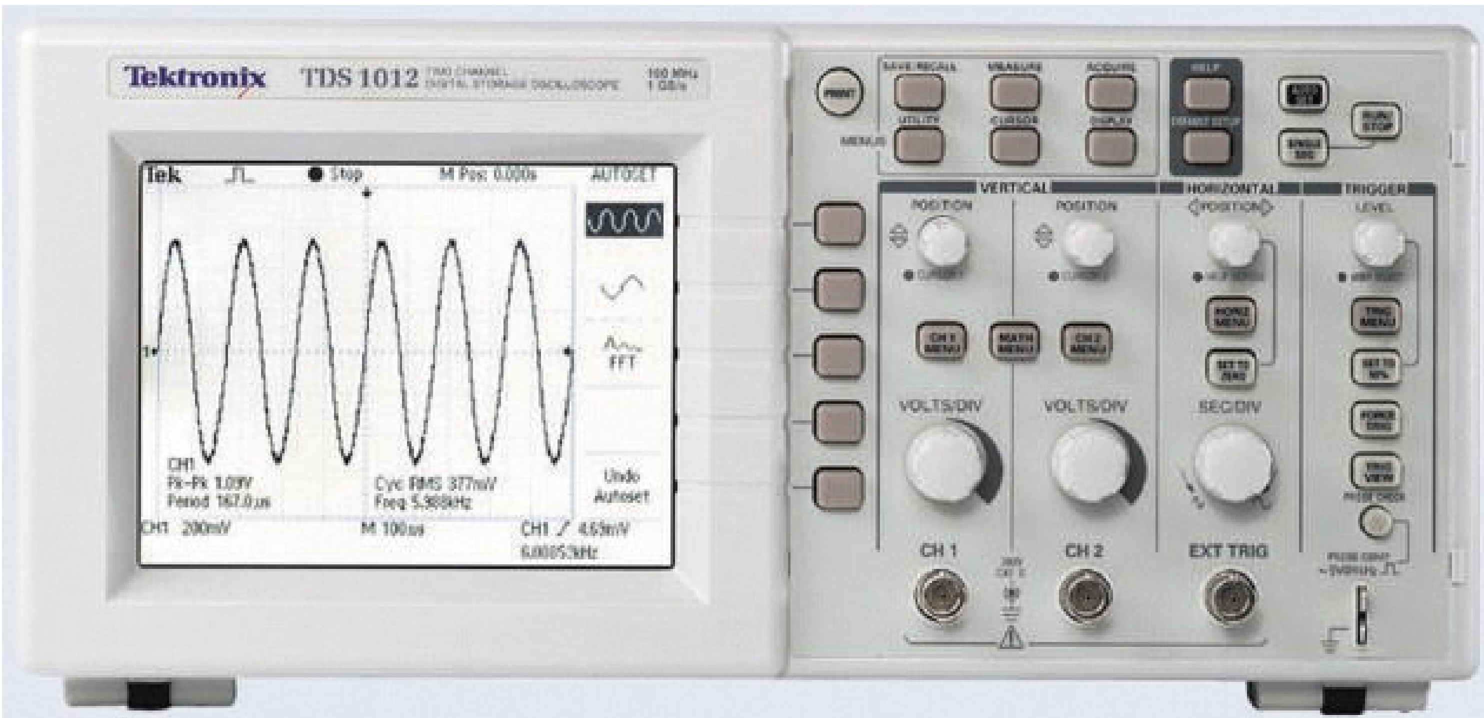
- Digital oscilloscopes use an analogue-to-digital converter (ADC) and appropriate processing.



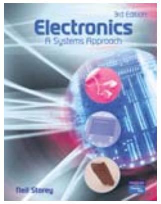
A simplified block diagram

Digital oscilloscope (contd.)

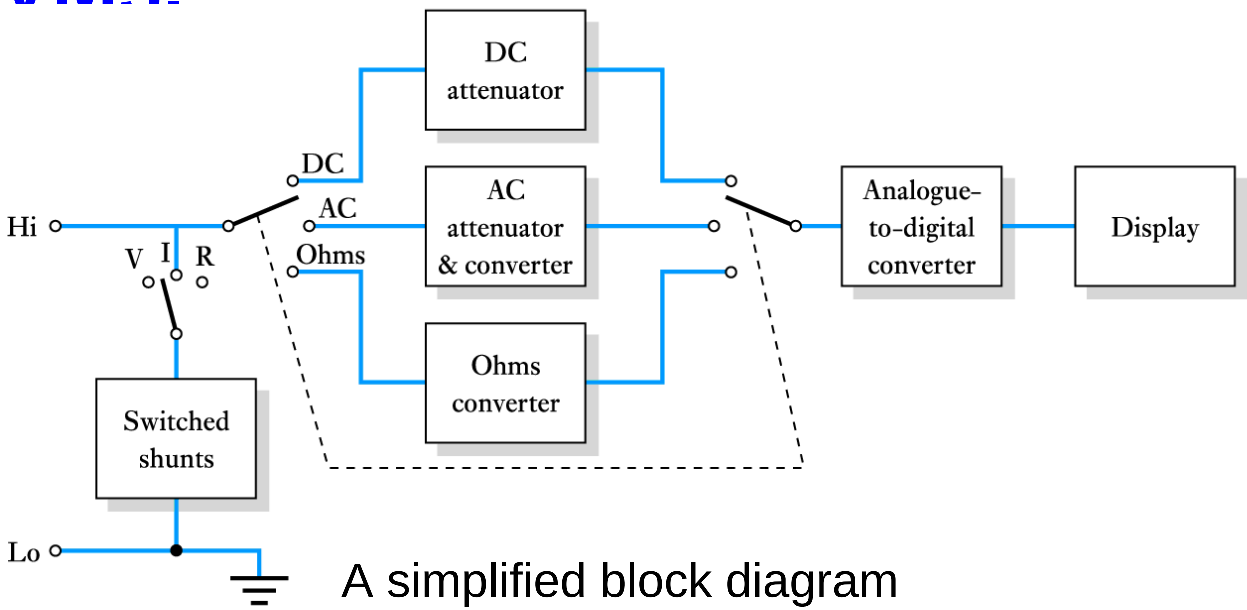
- A typical digital oscilloscope



Digital multimeters



- **Digital multimeters** (**DMMs**) are often (inaccurately) referred to as **digital voltmeters** or **DVMs**.
  – A
  c



A simplified block diagram

Digital multimeters (contd.)

- Measurement of voltage, current and resistance is achieved using appropriate circuits to produce a voltage proportional to the quantity to be measured.
  - In simple DMMs alternating signals are rectified as in analogue multimeters to give their average value which is multiplied by 1.11 to directly display the r.m.s. value of sine waves.
  - More sophisticated devices use a **true r.m.s. converter,** which accurately produces a voltage proportional to the r.m.s. value of an input waveform.

A typical digital multimeter

Key points

- A wide range of sensors is available.
- Some sensors produce an output voltage related to the measured quantity and therefore supply power.
- Other devices simply change their physical properties.
- Interfacing may be required to produce signals in the correct form.
- Most actuators take power from their inputs in order to deliver power at the output – the efficiency is often low.
- We often sense quantities in order to measure them – there are a number of standard measuring instruments.

Repetition

- Ohms lag

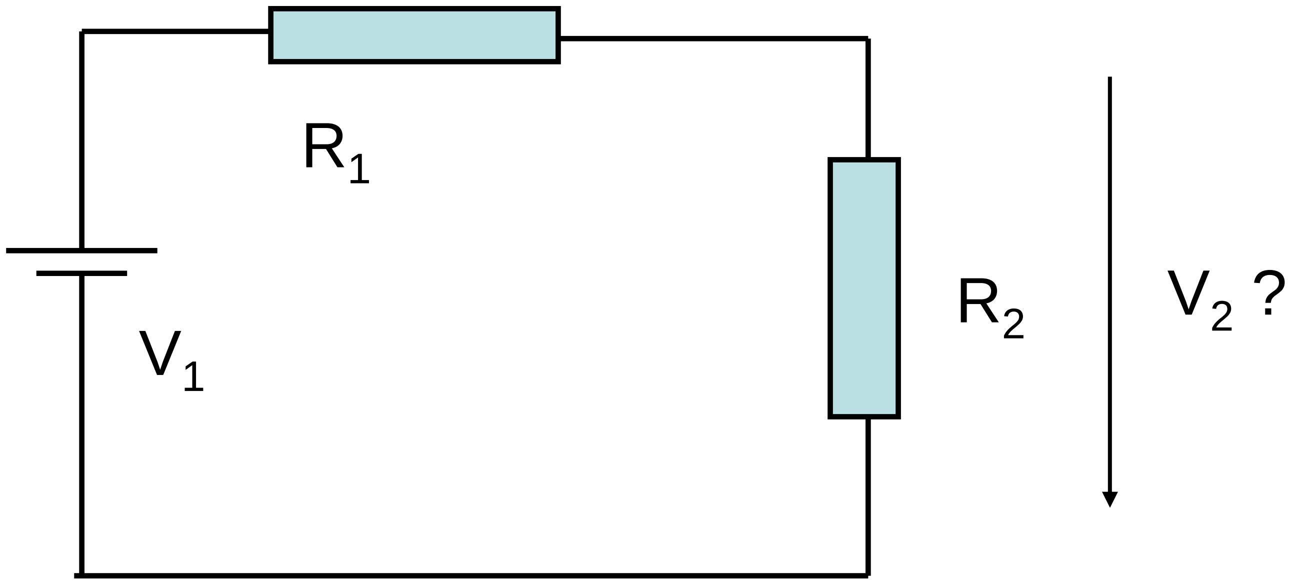
  $V = R\,I$

- Kirchhoffs lagar

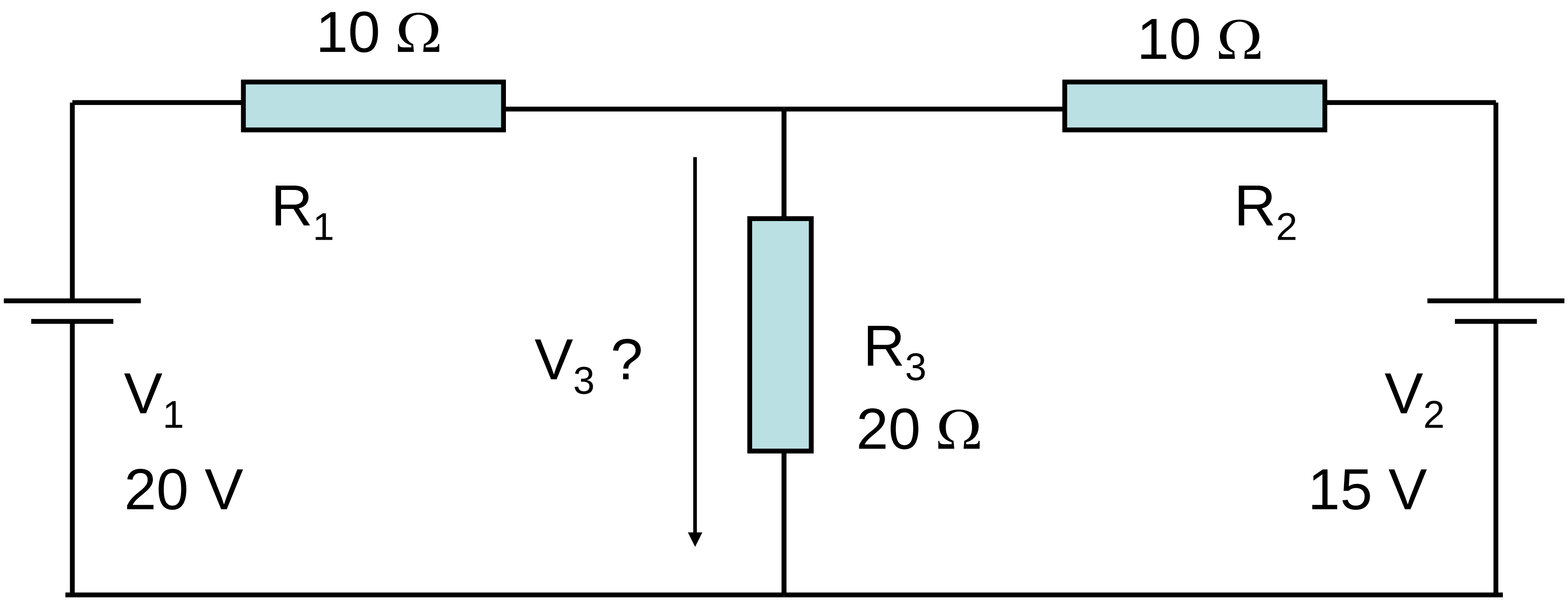
  $\Sigma_i I_i = 0$ (summan av strömmarna i en nod lika med noll)

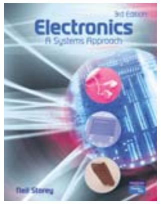
  $\Sigma_i V_i = 0$ (summan av spänningsfall i en sluten slinga lika med noll)

Exempel



Exempel 1.11

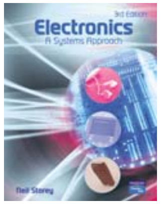




Amplification

- **Introduction**
- **Electronic amplifiers**
- **Sources and loads**
- **Equivalent circuit of an amplifier**
- Thévenin and Norton equivalent circuits
- Output power
- Power gain
- Frequency response and bandwidth
- Low- and high-frequency effects
- Noise
- Differential amplifiers
- Simple amplifiers.

Introduction

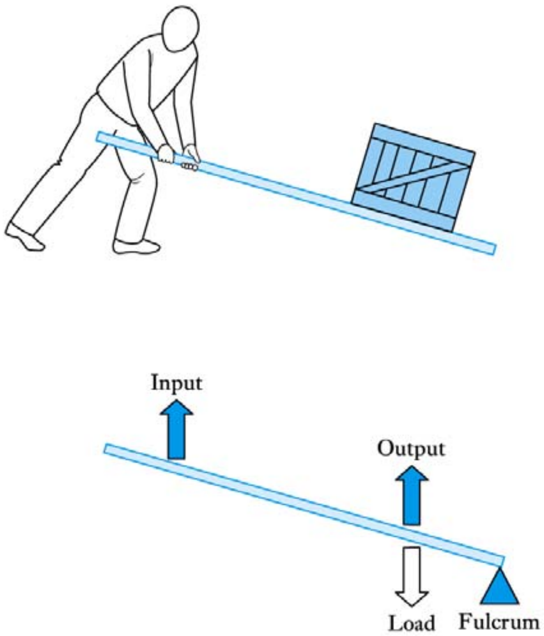


3.1

- Amplification is one of the most common processing functions.
- **Amplification** means making things bigger.
- **Attenuation** means making things smaller.
- There are many non-electronic forms of amplification.

Amplifiers

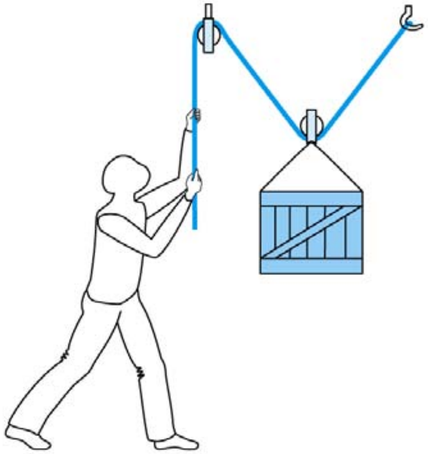
- **Non-electronic amplifiers**
  - **Levers**
    - Example shown on the right is a force *amplifier*, but a displacement *attenuator.*
    - Reversing the position of the input and output would produce a force *attenuator* but a displacement *amplifier.*
    - This is an example of a **non-inverting amplifier** (since the input and output are in the same direction).

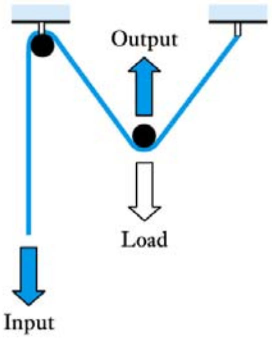


A lever arrangement

Amplifiers (contd.)

- **Non-electronic amplifiers**
  - **Pulleys**
    - Example shown right is a force *amplifier*, but a displacement *attenuator.*
    - This is an example of an **inverting amplifier** (since the input and output are in opposite directions) but other pulley arrangements can be non-inverting.





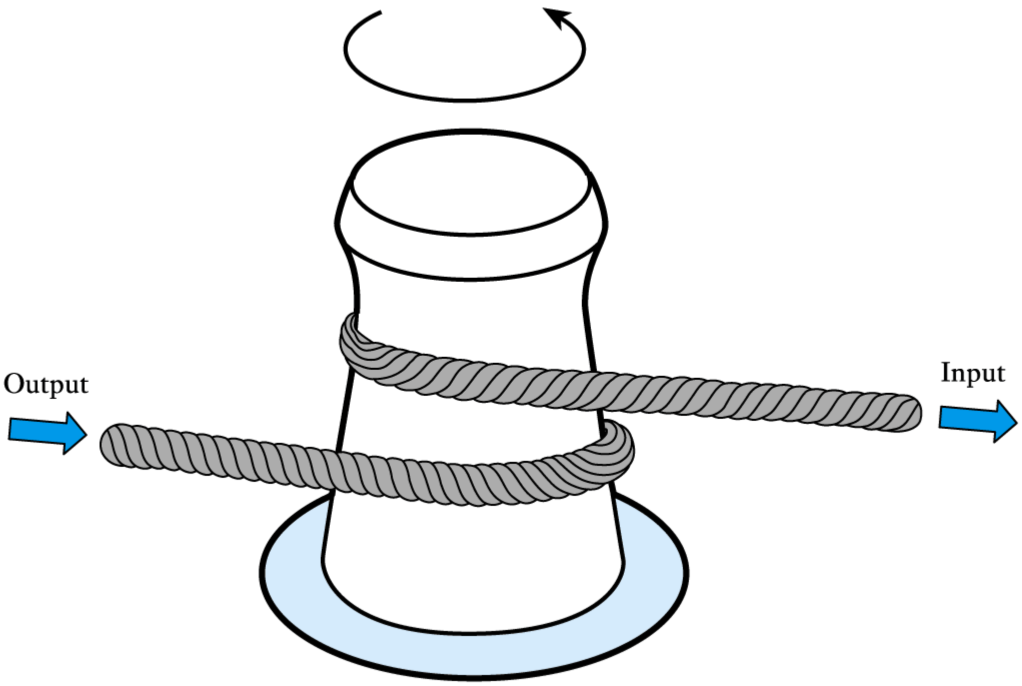
A pulley arrangement

Amplifiers (contd.)

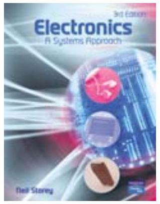
- **Passive and active amplifiers**
  - Levers and pulleys are examples of **passive amplifiers** since they have no external energy source.
    - In such amplifiers the power delivered at the output must be less than (or equal to) that absorbed at the input.
  - Some amplifiers are not passive but are **active amplifiers** in that they have an external source of power.
    - In such amplifiers the output can deliver more power than is absorbed at the input

Amplifiers (contd.)

- **Non-electronic active amplifiers**
  - an example is the torque amplifier shown here



A torque amplifier



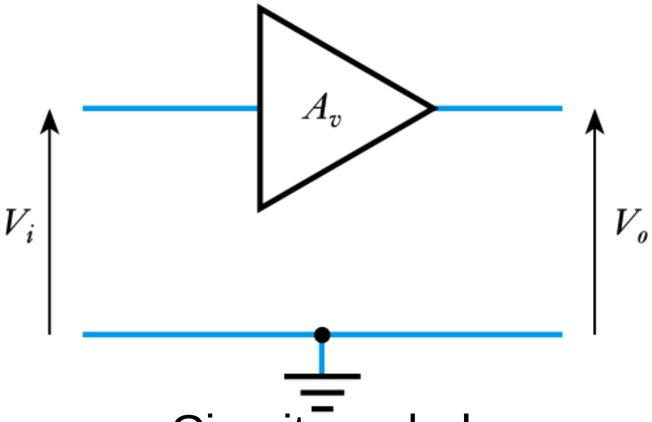
Electronic amplifiers

- Can be passive (e.g. a transformer) but most are active
- We will concentrate on *active* electronic amplifiers
  - take power from a power supply
  - amplification described by gain

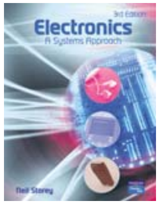
$$\text{Voltage Gain}\,(A_v) = \frac{V_o}{V_i}$$

$$\text{Current Gain}\,(A_i) = \frac{I_o}{I_i}$$

$$\text{Power Gain}\,(A_p) = \frac{P_o}{P_i}$$

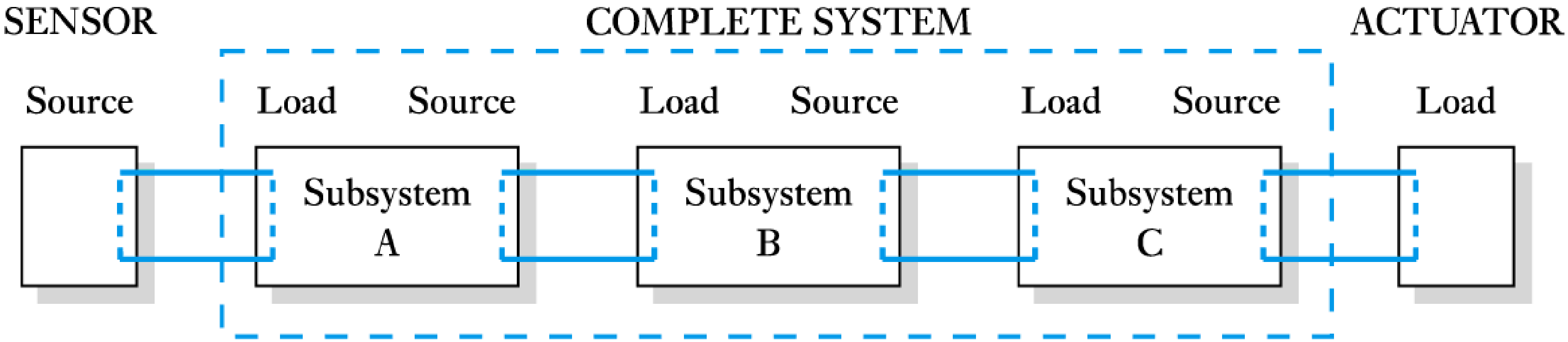


Circuit symbol



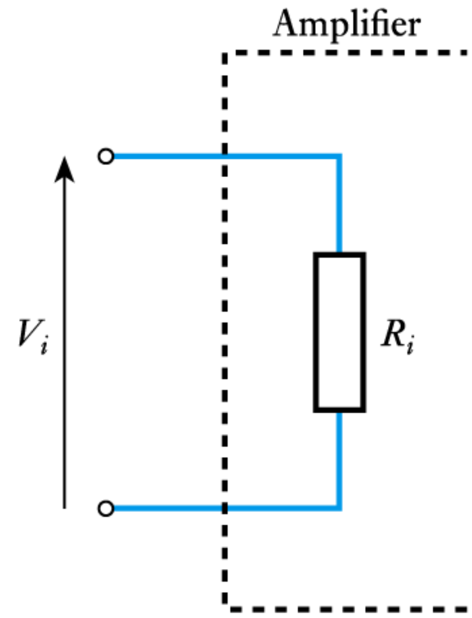
Sources and loads

- An *ideal* voltage amplifier would produce an output determined only by the input voltage and its gain.
  – irrespective of the nature of the source and the load
  – in real amplifiers this is not the case
  – the output voltage is affected by **loading.**



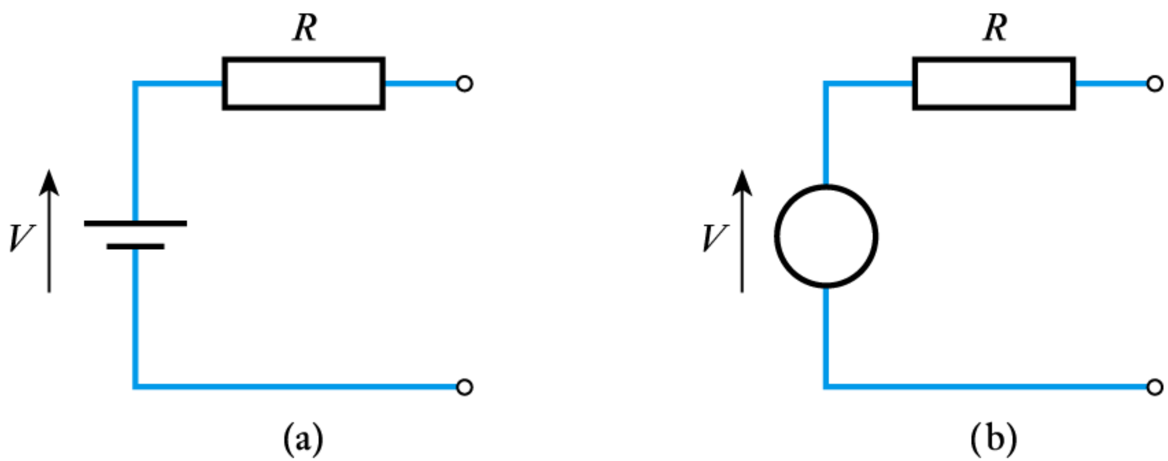
Modelling sources and loads

- **Modelling the input of an amplifier**
  - the input can often be adequately modelled by a simple resistor
  - the **input resistance.**



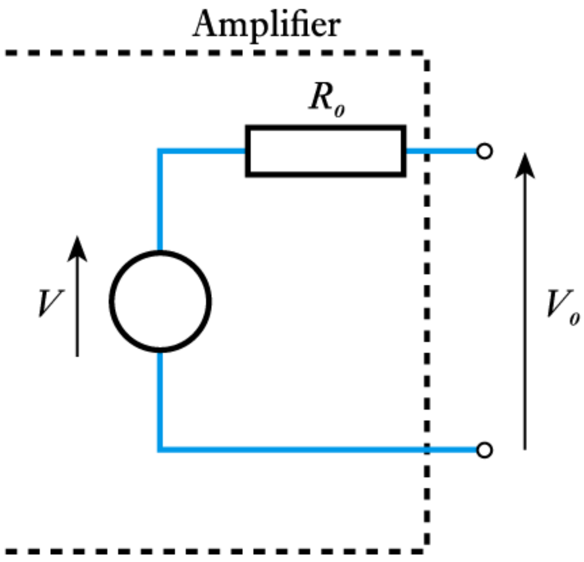
Modelling sources and loads (contd.)

- **Modelling the output of a circuit**
  - all real voltage sources have an output resistance
  - for example, a battery can be represented by an ideal voltage source and a series resistance representing its **output resistance.**



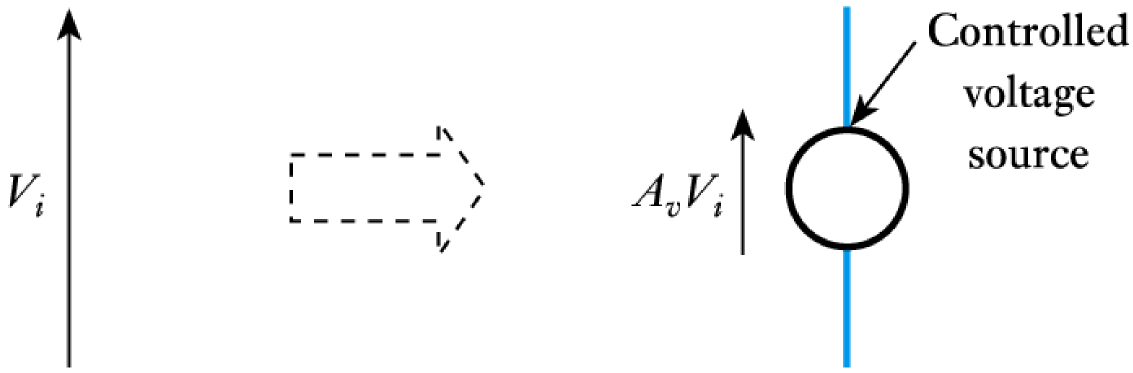
Modelling sources and loads (contd.)

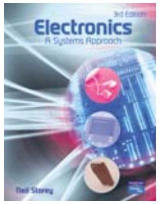
- **Modelling the output of an amplifier**
  – Similarly, the output of an amplifier can be modelled by an ideal voltage source and an output resistance.
  – This is an example of a **Thévenin equivalent circuit** (we will return to such circuits later).



Modelling sources and loads (contd.)

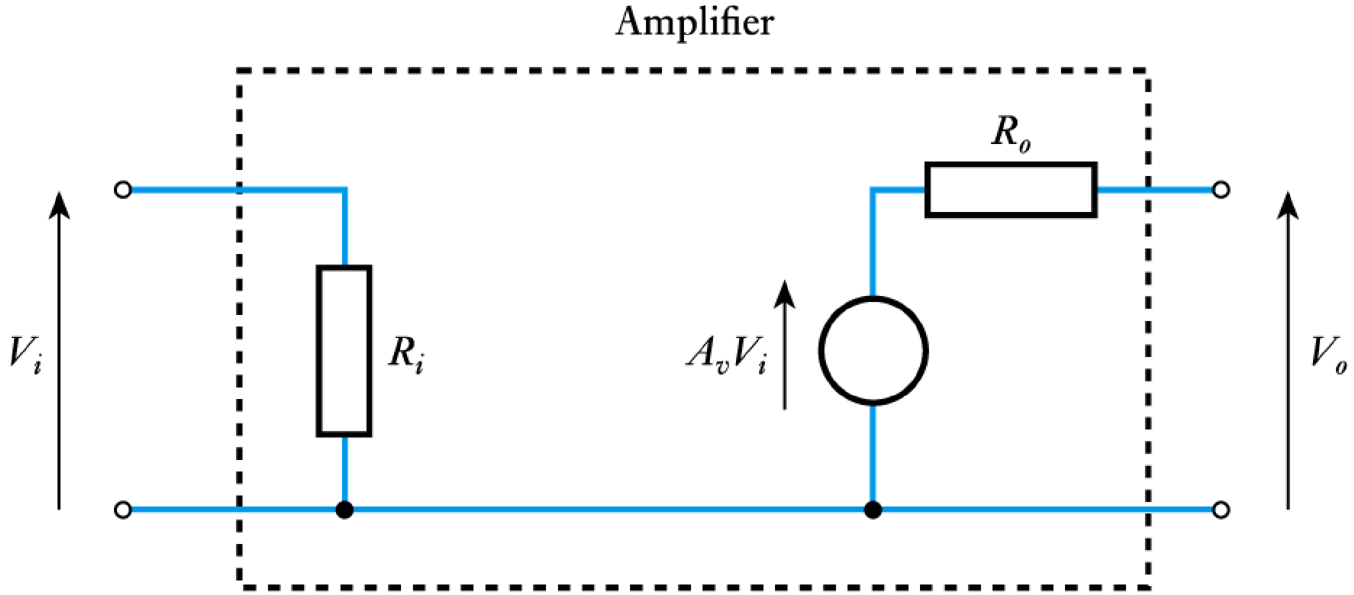
- **Modelling the gain of an amplifier**
  - can be modelled by a controlled voltage source
  - the voltage produced by the source is determined by the input voltage to the circuit.





Equivalent circuits of amplifiers

- Having modelled the input, the output and the gain, we can now model the entire amplifier.



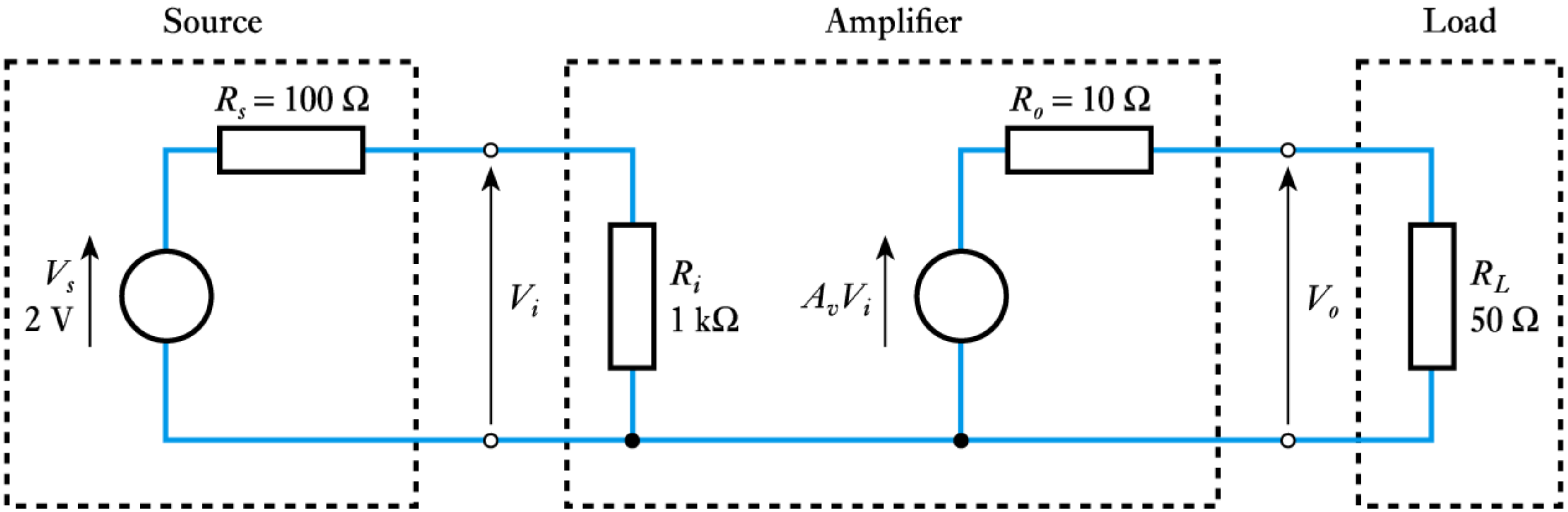
Equivalent circuits of amplifiers (contd.)

- **The use of an equivalent circuit**
  (see **Example 3.1** in the course text):

  Example: An amplifier has a voltage gain of 10, an input resistance of 1 kΩ and an output resistance of 10 Ω. The amplifier is connected to a sensor that produces a voltage of 2 V and has an output resistance of 100 Ω, and to a load of 50 Ω. What will be the output voltage of the amplifier (that is, the voltage across the load resistance)?

Equivalent circuits of amplifiers (contd.)

- We start by constructing an equivalent circuit of the amplifier, the source and the load.



Equivalent circuits of amplifiers (contd.)

- From this we can calculate the output voltage:

$$V_i = \frac{R_i}{R_s + R_i} V_s$$

$$= \frac{1\text{k}\Omega}{100\,\Omega + 1\text{k}\Omega} 2\text{ V} = 1.82\text{ V}$$

$$V_o = A_v V_i \frac{R_L}{R_o + R_L}$$

$$= 10\ V_i \frac{50\ \Omega}{10\ \Omega + 50\ \Omega}$$

$$= 10 \times 1.82\ \frac{50\ \Omega}{10\ \Omega + 50\ \Omega} = 15.2\text{ V}$$

Equivalent circuits of amplifiers (contd.)

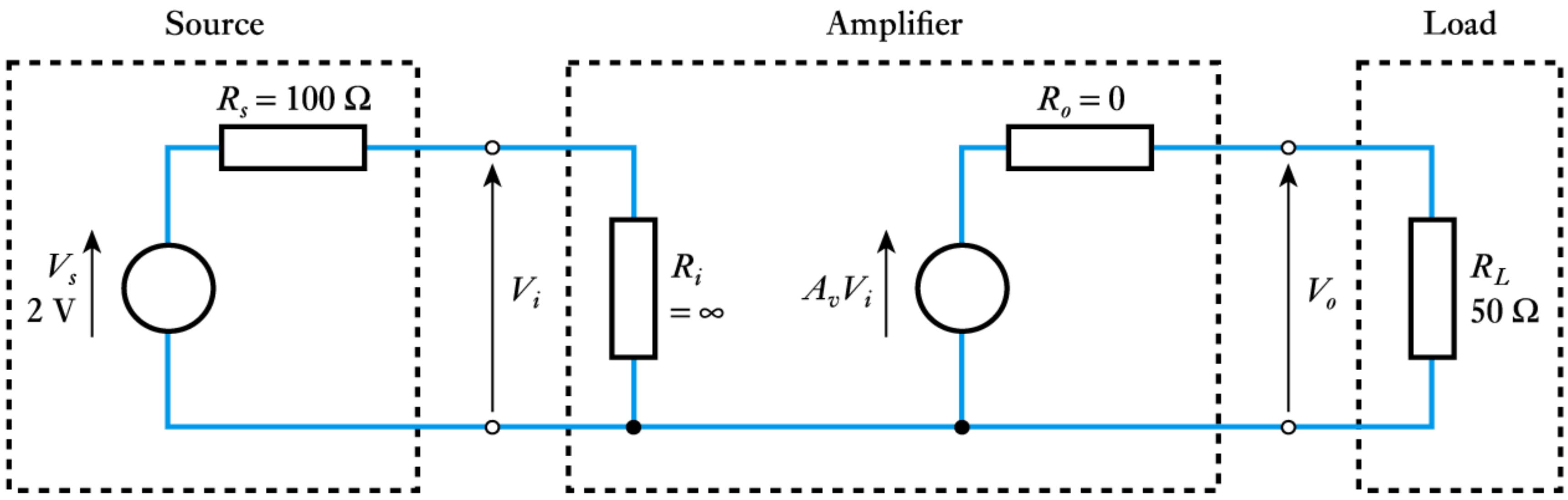
- The **voltage gain** of the circuit in the previous example is given by:

$$\text{Voltage gain } (A_V) = \frac{V_o}{V_i} = \frac{15.2}{1.82} = 8.35$$

  - note that this is considerably less than the stated gain of the amplifier (which is 10)
  - this is due to **loading effects**
  - the gain of the amplifier in isolation is its **unloaded voltage gain.**

Equivalent circuits of amplifiers (contd.)

- An **ideal voltage amplifier** would not suffer from loading
  - it would have $R_i = \infty$ and $R_o = 0$
  - consider the effect on the previous example.



Equivalent circuits of amplifiers (contd.)

- If $R_i = \infty$, then $\dfrac{R_i}{R_s + R_i} \approx \dfrac{R_i}{R_i} = 1$
- Therefore

$$V_i = \frac{R_i}{R_s + R_i} V_s \approx V_s = 2\ \mathrm{V}$$

$$\begin{aligned} V_o &= A_v V_i \frac{R_L}{R_o + R_L} \\ &= 10\ V_i \frac{50\ \Omega}{0\ \Omega + 50\ \Omega} \\ &= 10 \times 2 \frac{50\ \Omega}{50\ \Omega} = 20\ \mathrm{V} \end{aligned}$$

– the effects of loading are removed (see **Example 3.3**).